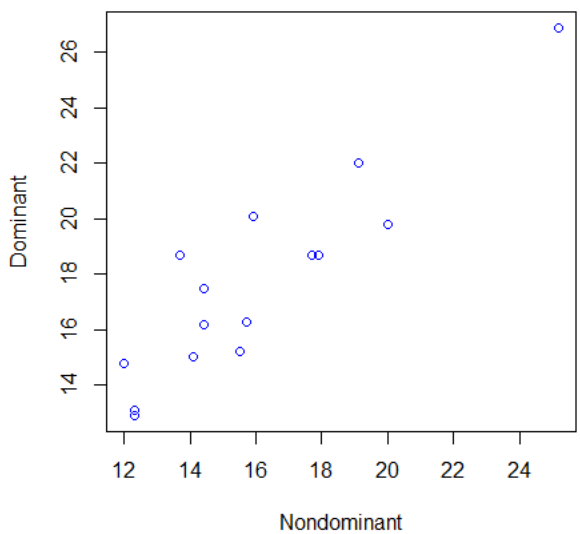
See R basic source file (Appendix) for computation details. You are not required to use R. You should note the tools that you use, and include source code either as an appendix or separate file. Source code is not graded in this course, but if your results are wrong I may be able to identify the cause. The discipline of writing clear (annotated) code is important to develop.

Task 1: Bone strength MMC (Ed 10) 2.14, 2.15, 2.16

2.14 **5+2+2+2+2** **13 points**

(a) Scatterplot



I receive a C for this graph. There is no title; the axis labels are not self-explanatory; and there are no measurement units. ☹

(b) **Overall pattern & striking deviations** Approximately linear, increasing (higher Nondominant value associated with higher Dominant value); one outlying point (top right); dominant arm strength is typically greater than non-dominant arm strength.
If the outlying point is masked, the remaining points are still positively associated, but the sense of linearity is reduced…there might be some curvature…?
Less obvious Two of the cases have dominant arm strength *less than* the nondominant arm strength: observations 6 (20.0, 19.8) Nm/1000, and 12 (15.5,15.2) Nm/1000. A grid on the scatterplot or showing the line of equality would reveal that more readily…or a "y=x" line that some of you sensibly included.

(c) **Form, direction, strength** Form and direction we covered in (b). Strength: there is not much scatter about a perceived straight line, compared with the y-range of the date overall…so the relationship appears moderately strong. We could formalize the linear strength measure with the correlation coefficient: 0.9 here.

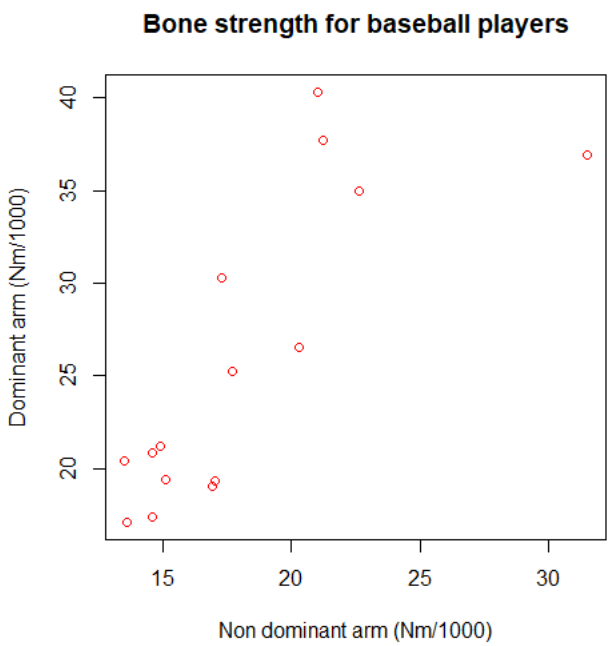
(d) **Outliers** There is a single outlier: in both the x and y directions (upper right of graph: coordinate 25.2Nm/1000, 26.9Nm/1000)…and appears to be consistent with a line we would draw through the main body of points.

(e) **Is the relationship approximately linear?** The relationship IS approximately linear – but see the caveat about possible curvature noted in (b).

Note: If you want to identify the values of the outlying point (usually a good idea), remember to give the measurement units. In a simple data set like this, the location of the outlier is easy to find: "upper right"; in other situations it may be more difficult. It is therefore a good idea to always explicitly identify such points with their values.

2.15 **5+2+2+2+2** **13 points**

(a) Scatterplot



Labeling much better…

(b) **Overall pattern** Approximately linear, increasing (higher Nondominant value associated with higher Dominant value); one outlying point – but only outlying in the x (Nondominant strength) direction. Notice that for these trained athletes the dominant arm is always stronger than the nondominant arm. There might be three mini-groups [can you identify these?]…but see the last sentence in (e) below…

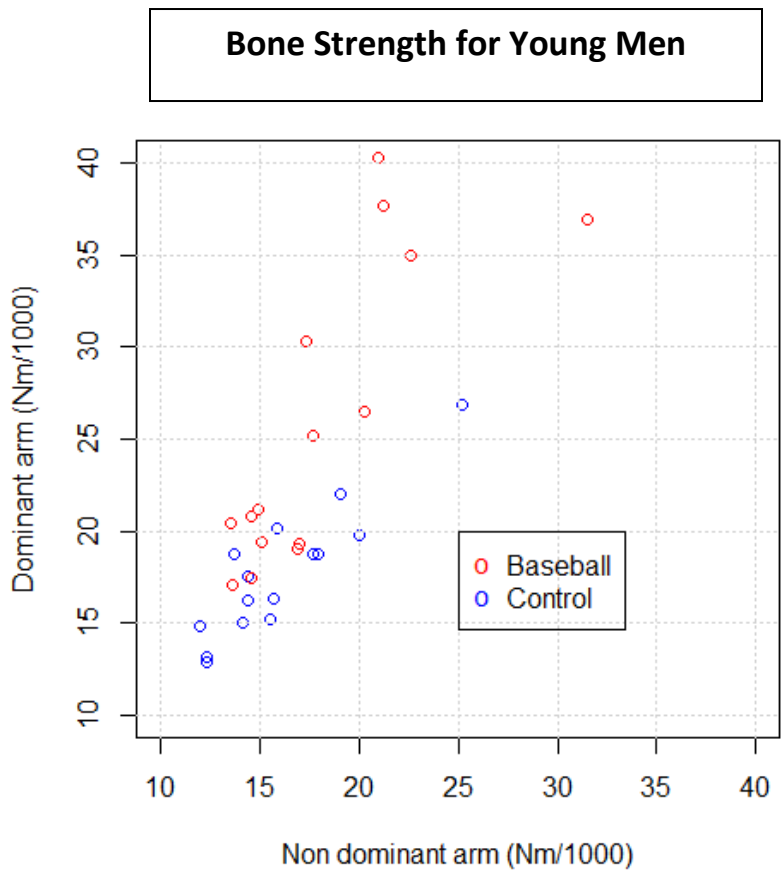
(c) **Form, direction, strength** Linear, positive association; the outlier is not on the linear relationship we eyeball for the other data: the strength of the dominant arm falls short for the unusually strong non-dominant arm. How strong is the linear association here? Without computing the correlation coefficient (done below) we see that the scatter is...moderate? Judgements differ (that is why we move on from description to quantification).

(d) **Outliers** There is a single outlier (nondominant value > 30 Nm/1000): in the x direction only…

(e) **Is the relationship approximately linear?** Overall, yes (excluding that outlying point, and noting the possibility of groups) – unlike the previous data, there is no hint of curvature here. *Some of you suggest that there is an exponential form (excluding the outlier) where the slope is flatter at low arm strengths and increases a little as arm strength increases. This can be tested by transforming the data and/or fitting suitable models and assessing estimated parameter significance…and thinking about what bio-mechanical structures might be present. One might also gather more data to see whether the eye is being drawn to something that is not real…Investigating that outlying point is another job that should be undertaken. Was there a transcription error? Is the player ambidextrous? Are there other baseball players with such strong non-dominant arms?*

2.16 Compare the two groups: Controls vs Baseball Players **5 + 5 10 points**

(a) scatterplot



I used color rather than symbol type to distinguish the two data sets. With the overlapping of points in the lower left area, color does a superior job in distinguishing the two sets.

(b) 'Reds' (baseballers) higher than 'blues' (controls) on the Dominant arm: baseballers throwing arm is stronger than in the control group.

Range of nondominant: Controls 12-20 (Nm/1000) vs Baseballers 14-23 (Nm/1000) – so the nondominant arms are a little stronger in the baseballers too…as if the training for the dominant arm has pulled along the strength of the nondominant arm…or naturally stronger people take up baseball…or …

The outlier for the baseballers lies on the eyeballed linear relationship for the controls…curious. If that relationship for controls does extend (there are no controls with those high strength measures), then the baseball outlier becomes an oddity in that there is no increased strength of the dominant arm…as if a naturally very strong person did not train? Or, the other way around: both arms trained…? Be wary here of making up a story for what could be a simple recording error.

The slope of eyeballed line is steeper for baseballers than controls. The training piles on the strength of those dominant arms.

Variability of controls (blue) about the linear trend is less than the variability of the baseballers (red) about their trend. *This can be quantified by fitting straight lines, but we don't need that here. See below.*

Task 2: Bone Strength (continued) 2.35, 2.36 10 points

2.35
(a) Correlation of the two variables is 0.90.
(b) Yes, the correlation is a good summary of the scatterplot (we saw this above): the relationship is linear with a single outlier that is consistent with the linear pattern of the main body of data.

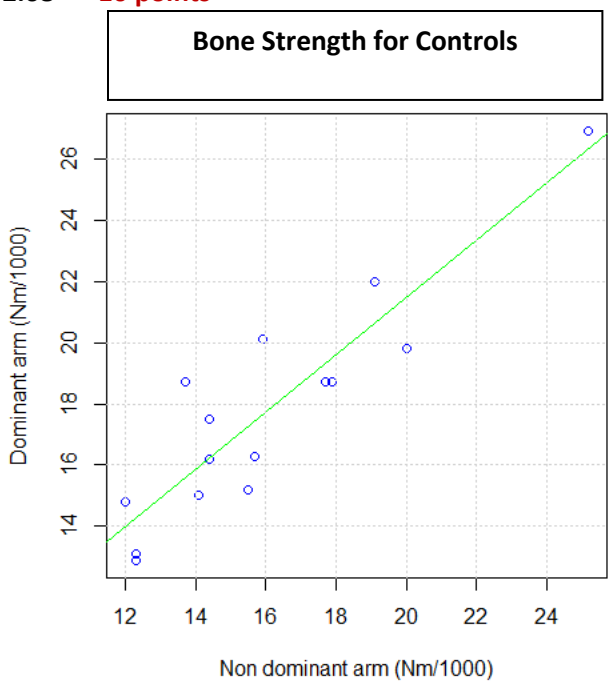
2.36
(a) Correlation of the two variables is 0.79.
(b) The correlation is only a reasonable summary of the scatterplot in this case – because of that outlier. Excluding the outlier, the correlation is 0.87. So, other than the outlying point, the correlation is a good summary.

Correlation is a “good” summary of two variables if the description “randomly scattered about a straight line” properly and entirely describes the data. It does not matter if the amount of scatter is tiny (correlation almost 1), or large, correlation 0.6 (say). What matters is if the variation has any pattern about an average straight line. Correlation cannot say anything about the presence of outliers, curvature, or any other non-random scatter about an average straight line.

Omitting data
We don't throw away data, but we can analyze subsets for understanding. And "outlier" means "Go take a careful look at that data...it may be the most interesting point in the set..."

Task 3 (NOT FOR GRADING) Bone strength (continued) 2.52 – 2.56

2.68 **10 points**



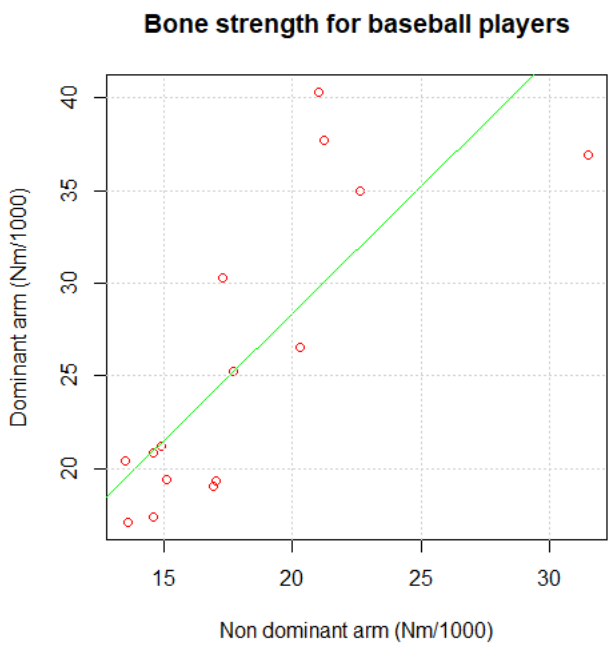
The commentary we would add here essentially confirms the eyeball remarks made above. In particular, notice how consistent the outlier is with the overall linear fit. The question is, if that point were not included in the fitting, how would the line look?

The line fitted here, to all the data, is $dominant = 2.74 + 0.936 \times nondominant$

The slope coefficient is less than one, so the bone strength of the dominant arm increases by less than one unit for each unit increase in the strength of the non-dominant arm. Is that surprising? Remember that this data is for a collection of people; it is not a description of what happens to a single person as that person builds strength. There are lots of reasons that might contribute to an explanation. We could begin by thinking about extremes. At ordinary levels of strength (what is that?) we might expect dominant arms to be stronger than non-dominant arms. At very high levels of strength we might expect both arms to have the same strength: both arms trained, or some biological limitation? Should we expect strength differences to be proportional or fixed? Lots of possibilities...

One of those possibilities is that we are over interpreting an outcome of sampling variability. There are just 15 observations here.

2.69 Same again for the baseball players **10 points**



The pulling effect of the outlier can be seen here…If the outlier is excluded, the fitted line will have a steeper slope and a much better representation of the body of the data…a good additional exercise for you.

The fitted line here is $dominant = 0.886 + 1.373 \times nondominant$

Immediately we notice that the slope coefficient is greater than one. For the baseballers, the dominant arm bone strength increases at a greater rate than does the non-dominant arm. Once again, why? Remember, this is not a set of data showing the effect of training on an individual. It is a set of measurements on baseball players. However, as a whole, it seems that training increases dominant arm strength by more than it increases non-dominant arm bone strength – if we are assuming that training is the fundamental cause of the differences. Perhaps naturally talented baseball players are also naturally quite asymmetric…

2.70 Predict the dominant arm strength for a control with non-dominant arm strength 16 cm^4/1000

This is just a matter of substituting the value 16 for x in the fitted line:

$$\begin{aligned} dominant &= 2.74 + 0.936 \times nondominant \\ &= 2.74 + 0.936 \times 16 \\ &= 17.716\ cm^4\ /1000 \end{aligned}$$

In the regression course, M644, a more detailed examination of model estimation and prediction is undertaken, and with more general models than this simple straight line model. The qualitative questions suggested above can be formally addressed with statistical analysis. "Is the slope coefficient for controls different from baseballers?", for example. It looks as though it is – can that conclusion be justified statistically?

2.71

$$\begin{aligned} dominant &= 0.886 + 1.373 \times nondominant \\ &= 0.886 + 1.373 \times 16 \\ &= 22.854\ cm^4/1000 \end{aligned}$$

2.72

(a) The difference in the predictions is $22.854\ -\ 17.716\ =\ 5.138\ cm^4/1000$
The revised text asks for a comparison of means, which is summarized here (borrowed from Fernando Rios Leon because I realized late):

	Young		Baseball Payers	
	Dominant	Non dominant	Dominant	Non dominant
Mean	16.01	17.73	18.12	25.77

We can observe that the mean among dominant values are almost same, but among Non-dominant values there is a big difference, the mean increase from 17.73 to 25.77

(b) The difference between the two groups of people (relevant here) is that the baseball players train while the controls do not. So the arm strength increase is a function of the explicit training. We must also remember that baseball players, being selected through a long process, are certainly more "athletic" on average than the general population, so we would expect strength measures to be greater even without the training. The higher slope coefficient of the baseball players model fit indicates that the increase in dominant arm strength for a one unit increase in non-dominant arm strength is considerably greater for baseball players than controls. The much lower intercept is also interesting…but it does not make sense to extrapolate (and certainly not to 0!) for the reasons in the text.

(c)

```
     nondom      dom    domB   diff
[1,]   12      13.972 17.362 3.390
[2,]   16      17.716 22.854 5.138
[3,]   20      21.460 28.346 6.886
```

(d) We see that the difference between baseball players and controls increases as the non-dominant arm strength increases (this reflects that higher slope coefficient).

APPENDIX Basic R code for the data analyses


```
##    !!! Code is for pedagogic purposes...Lots of scope for improvement
##    !!! efficiency, error checking, ...and use of the more advanced
##    !!! packages such as ggplot...
##
## ---------------------------------------------------------------------------

## There is limited commentary here; refer to the exercises in the text
## for details.
##      AND USE THIS AS AN OPPORTUNITY TO ADD YOUR OWN CODE COMMENTARY

## ============================= Bone strength

## ===== 2.24

Nondominant = c(15.7, 25.2, 17.9, 19.1, 12.0, 20.0, 12.3, 14.4,
                15.9, 13.7, 17.7, 15.5, 14.4, 14.1, 12.3)

Dominant    = c(16.3, 26.9, 18.7, 22.0, 14.8, 19.8, 13.1, 17.5,
                20.1, 18.7, 18.7, 15.2, 16.2, 15.0, 12.9)

## (a) scatterplot

plot(x=Nondominant, y=Dominant, col='blue')


## (b) describe overall pattern; any striking deviations

## (c) describe the form, direction, and strength of the relationship

## (d) identify any outliers

## (d) is the relationship approximately linear


## ===== 2.25 Bone strength for baseball players

NondominantB = c(17.0, 16.9, 17.7, 21.2, 21.0, 14.6, 31.5, 14.9,
                 15.1, 13.5, 13.6, 20.3, 17.3, 14.6, 22.6)

DominantB    = c(19.3, 19.0, 25.2, 37.7, 40.3, 20.8, 36.9, 21.2,
                 19.4, 20.4, 17.1, 26.5, 30.3, 17.4, 35.0)

## (a)

plot(x=NondominantB, y=DominantB, col='red',
  xlab='Non dominant arm (Nm/1000)',
  ylab='Dominant arm (Nm/1000)')
title("Bone strength for baseball players")
```

```
## ===== 2.26 Compare the baseball players with the controls


## (a) plot dall data on single graph, distinguishing the two
##     sets by plot symbol

plot(x=Nondominant, y=Dominant, col='blue', xlim=c(10,40), ylim=c(10,40),
  xlab='Non dominant arm (Nm/1000)',
  ylab='Dominant arm (Nm/1000)')
title("Bone strength for baseball players")
points(x=NondominantB, y=DominantB, col='red')
grid()
legend(25,20,legend=c("Baseball","Control"),
  pch=c('o','o'),col=c('red','blue'))




## ===== 2.47

## (a) correlation

cor(Dominant, Nondominant)

## ===== 2.48

## (a) correlation

cor(DominantB, NondominantB)

noOutlier = NondominantB < 30
cor(DominantB[noOutlier], NondominantB[noOutlier])



## ===== 2.68

## (a) scatter plot done previously
## (b) add given regression line

plot(x=Nondominant, y=Dominant, col='blue',
  xlab='Non dominant arm (Nm/1000)',
  ylab='Dominant arm (Nm/1000)')
title("Bone strength for baseball players")
# here we fit the model and draw the fitted line;
abline(lm(Dominant ~ Nondominant), col='green')
grid()

# if you want to draw the line directly from the coefficient estimates
# given in the text:

xpt = 10:30
ypt = 2.74 + 0.936*xpt
lines(xpt, ypt)

## (c)
```

```
## ===== 2.69
## (a) scatter plot done previously
## (b) add given regression line

plot(x=NondominantB, y=DominantB, col='red',
  xlab='Non dominant arm (Nm/1000)',
  ylab='Dominant arm (Nm/1000)')
title("Bone strength for baseball players")
# here we fit the model and draw the fitted line;
abline(lm(DominantB ~ NondominantB), col='green')
grid()


## ===== 2.70

2.74 + 0.936*16


## ===== 2.71

0.886 + 1.373*16


## ===== 2.72

## (a)
22.854 - 17.716

## (b)
## (c)

nondom = c(12,16,20)
dom    = 2.74  + 0.936*nondom
domB   = 0.886 + 1.373*nondom

# table:
cbind(nondom,dom,domB,domB-dom)

## (d)
```